MAHATMA GANDHI UNIVERSITY , KOTTAYAM

MGU-UGP(HONOURS)REGULAR EXAMINATION , OCTOBER 2025

THIRD SEMESTER

MG3 DSC PHL202: Introduction to Logic

QP CODE:25900546

PART A

1. True premise and conclusion – argument
2. Informative – express information or declarative form - emotive- expression of emotions
3. All men are Mortal
   All politicians are liars
4. False
5. Sita is clever and tall
   Sc.t
   Raju is bright and smart
   Rb.s
   . is constant

   P,q r ,etc are variables
6. Rb.s
7. P **≡ ~~p**
   p **→q is logically equivalent to ~q →~p**
8. p **→q**
   p^q
9. A statement that is always false
10. 1. A.B/ ∴A.B
    2. A (1,SIMP)
    3.AVC(2,Add)

11. modus tollens

12. systematic explanation of the relation between the premises and the conclusion. No scope for ambiguity

PART B

13. precise in nature- no scope of ambiguity – systematic explanation

14. predicate logic- analyse internal structure-object and relation – use predicate symbols

Propositional logic- sentences as atomic units

15

p	q	p↔q
T	T	T
T	F	F
F	T	F
F	F	T

16. if p is true

p. ~p if false

if p is false

p.~p is false

17. (p V q) → (p → ~q)

T T T F T F F T

T T F T T T T F

F T T T F T F T

F F F T F T T F

B) (P → Q) → (P V q)

T T T T T T T

T F F T T T F

F T T T F T T

F T F F F F F

18. p q r [(P → q) . (q → r) . (p → r)] → ~ q

T T T T T T T T T T T T F F T

T T F T T T F T F F F T F F T F T

T F T T F F F F T T F T T T T T F

T F F T F F T F T F F T F F T T F

F T T F T T T T T T T F T T F F T

F T T F T T F T F F F F T F T F T

F F F F T F T F F T F F T F F T F

Invalid argument

19. (p → q) → (~ q → ~ p)

T T T T F T T F T

T T F F T F F F T

F F T T F T T T F

F T F T T F T T F

VALID argument

b) [(p → q) . (p V q)] → q

T T T T T T T T T

T F F F T T F T F

F T T T F T T T T

F T F F F F F T F

VALID ARGUMENT

20. A Single natural language argument can be symbolized in more than one way depending on how we analyze its sentences.same argument can symbolized differently – logical equivalence

21. constructive dilemma – affirms the consequent of two conditional statement

Destructive dilemma-- Denies the consequent of two conditional statement

PARTC

22. P → (P → q)

T T T T T

T T T T F

F T F F T

F T F T F

Not Tautology

b) p → (p V q)

T T T T T

T T T T F

F T F T T

F T F F F

Tautology

c) (p . p) → p

T T T T T

T T T T T

F F F T F

F F F T F

Tautology

23. language as a medium for the expression of logical thought- valid reasoning- symbolic logic remove ambiguities-

24. compound statement- connecting simple statemen with logical connectives- and, or, if.. then ,if and only if

25. p→(q.r)

∴~(q.r) →~p

Conditional form: If p implies (q.r), then not (q.r)implies not p

Contraceptive form (~(q.r) →~p) is the contraceptive of (p→(q.r))

26. Argument have multiple argument form – symbolized in logically equivalent way- keep validity also